

3.1.2 Custom PCB Fabrication (soltra-sensor-node-pcb)

While breadboards and development boards are sufficient for Monozukuri prototyping, the IDP commercialization phase necessitated the fabrication of custom Printed Circuit Boards (PCBs).

The physical foundation of the edge node is a custom-designed, single-sided Printed Circuit Board (PCB) fabricated on a standard FR4 substrate. The unpopulated board (illustrated in Figure 3.1.X) highlights the optimized trace routing required to securely interface the central Saeed Studio XIAO ESP32C3 with the external sensor payload.

Due to the design deliberately utilizes a single-sided copper layer to streamline manufacturing and reduce costs, the routing topology was meticulously planned to prevent signal crossover and maintain a clean ground return path without relying on complex internal vias. The layout features precisely aligned footprint pads for the I2C-based Adafruit TSL2591 irradiance sensor and the analog GUVVA-S12SD UV sensor. Furthermore, the trace design accommodates the integral battery voltage divider circuitry, ensuring clean analog signal integrity to the main microcontroller.

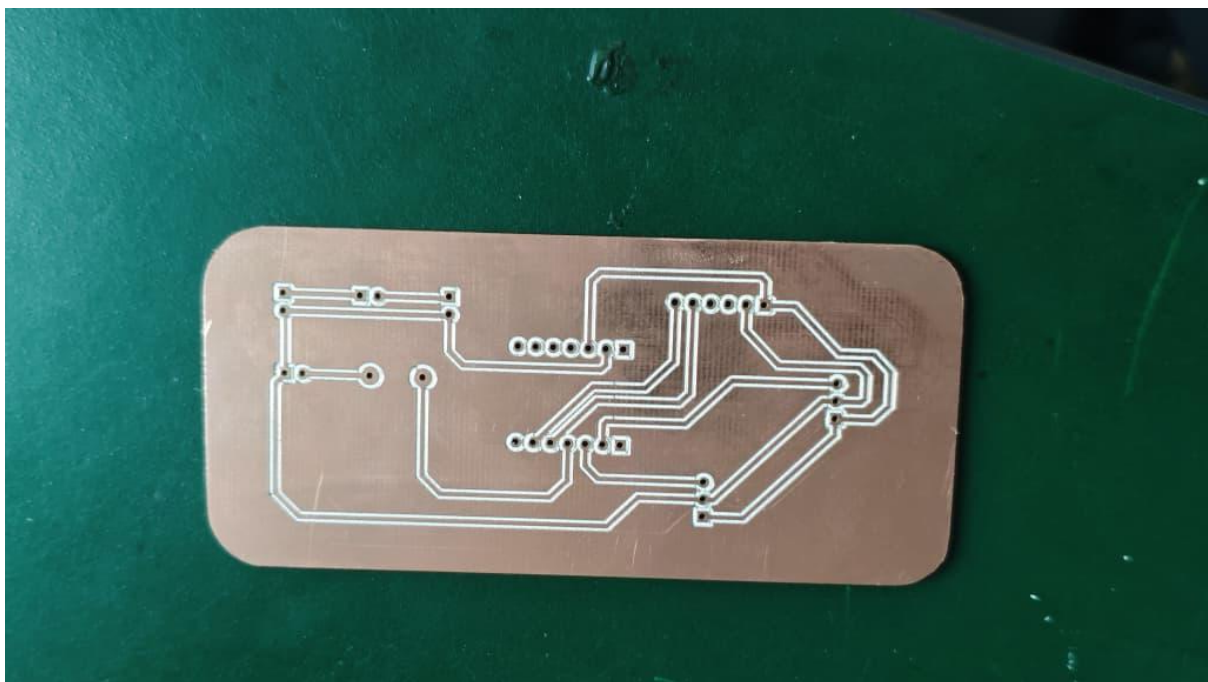


Figure 3.1.x the bare single-sided PCB prior to component assembly, showcasing the trace pathways, mounting holes, and component footprints.

3.1.3 The Heltec V3 ESP32-S3 Master Hub

The central brain of the Soltra ecosystem is the **Heltec WiFi LoRa 32 V3**. This specific development board was selected because it houses the immensely powerful **ESP32-S3** processor.

The S3 processor provides two massive advantages over traditional microcontrollers:

1. **Dual-Core FreeRTOS Topology:** The firmware isolates tasks aggressively. Core 0 handles the heavy TLS encryption required for the HiveMQ Cloud MQTT connection [3] and the Next.js HTTPS POST requests. Core 1 executes the Prioritized Finite State Machine (PFSM) [2], crunching the incoming edge node telemetry and performing the heavy floating-point trigonometry required for the Ephemeris solar position calculations.
2. **Vector Instructions (SIMD):** Essential for rapidly accelerating the complex math required for the internal solar tracking algorithms.

3.1.4 The ESP-NOW Auto-Pairing Protocol

Managing a dynamic wireless mesh network introduces significant onboarding complexities. If an edge node fails and is replaced, the system must recognize it without requiring the homeowner to manually input MAC addresses.

To solve this, the Soltra engineering team developed a zero-configuration auto-pairing protocol utilizing **ESP-NOW**. ESP-NOW is an Espressif-proprietary, connectionless Wi-Fi protocol that transmits raw MAC-to-MAC frames, bypassing the massive overhead (and pairing times) of standard TCP/IP router networks.

The Pairing Sequence:

1. Upon factory reset, a XIAO edge node enters promiscuous mode and rapidly scans Wi-Fi channels 1 through 13.
2. It continuously broadcasts a *PairingReqPkt* (containing a Magic Byte 0x99 and its specific *NODE_ID*).
3. The Master Hub, listening on its active channel, receives the broadcast. It registers the node's MAC address into its peer list and fires back a *PairingAckPkt* (Magic Byte 0xAA).

4. The edge node writes this pairing data into Non-Volatile Storage (NVS) using the `<Preferences.h>` library, allowing it to instantly transmit its *SensorPkt* payloads on all subsequent wake cycles without ever scanning again.

3.1.5 The ESP32-S3-CAM MJPEG Node

Beyond the sensor mesh, the Soltra hardware suite includes a dedicated **ESP32-S3-CAM** node (soltra-camera-node). Recognizing that remote operators often require physical visual confirmation of the panel's mechanical state, this node is mounted directly to the tracker's mast.

It generates a continuous, high-framerate MJPEG video stream over the local Wi-Fi network. This stream is intercepted by the desktop and mobile HUD applications, providing a live visual feed perfectly synchronized with the incoming MQTT numerical telemetry, allowing operators to safely monitor extreme weather events without physically stepping onto the site.

3.1.6 Printed Sensor Housing

To ensure the longevity of the decentralized edge nodes in harsh outdoor environments, a custom protective enclosure was engineered to house the Seed Studio XIAO ESP32C3 and the custom soltra-sensor-node-pcb. This is because the nodes are continuously exposed to intense thermal radiation and ultraviolet (UV) degradation, the housing was 3D printed utilizing Polyethylene Terephthalate Glycol (PETG) filament. PETG was selected over standard PLA due to its superior glass transition temperature and excellent inherent UV resistance, preventing the casing from warping or becoming brittle during peak daylight hours.

The enclosure design prioritizes sensory accuracy. The internal chassis is modeled with tight-tolerance standoffs to secure the PCB, preventing mechanical failure from wind-induced vibrations. To protect the delicate 3.3V logic components from moisture and dust ingress, the cable routing ports for the battery and sensor wires feature minimal clearance gaps. Meanwhile, the top face of the housing incorporates a strategically designed aperture to ensure the TSL2591 irradiance sensor, the analog UV sensor and the soltra-camera-node maintain an unobstructed field of view to the sky without exposing the core microcontroller to the elements.

Staying true to the Soltra ecosystem's core objective of being a frictionless retrofit solution, the housing utilizes a modular clamping mechanism. The main enclosure is structurally anchored to the mounting clamp utilizing robust M4 screws. This threaded mechanical fastening ensures

high shear strength, preventing any slippage or shifting of the sensor payload during the continuous dual-axis actuation of the solar array. Consequently, the edge nodes can be securely bolted onto the aluminium extrusion frame of virtually any existing static solar panel without requiring permanent physical modifications.

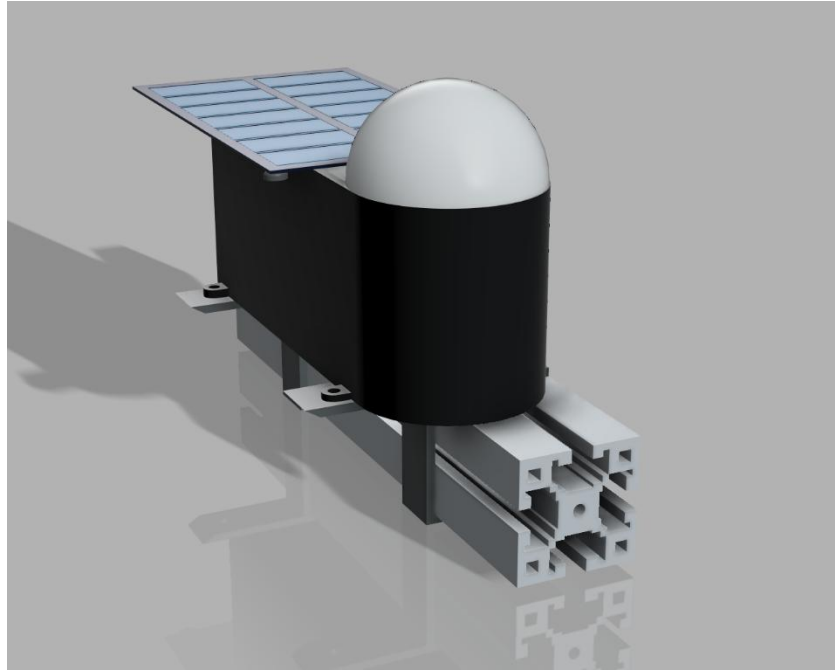


Figure 3.1.2 illustrates the high-fidelity Autodesk Fusion 3D render of the sensor housing, demonstrating the internal component mounting and external aerodynamic profile utilized for the final physical fabrication.

CHAPTER 9

SOLAR GUARDIAN TELEMETRY AND PEAK-WINDOW PERFORMANCE ANALYSIS

A core objective of the Soltra project was to validate the tracking efficiency and power generation of the AI-assisted dual-axis tracking system. Due to severe, sudden meteorological degradation (heavy monsoon cloud cover and storm conditions) restricting the collection of a full diurnal yield curve, this analysis focuses on a high-resolution micro-study. Data was captured during the most critical thermodynamic generation window: solar noon (12:10 PM to 1:00 PM).

To validate the system's performance, raw environmental data collected by the Soltra Sensor Nodes was comparatively analyzed against the actual electrical yield recorded by the MPPT (Maximum Power Point Tracking) charge controller, managed and logged via the Solar Guardian software suite.

9.1 PEAK IRRADIANCE & TRANSIENT TRACKING ANOMALY (12:30 PM - 1:00 PM)

During this 50-minute peak window, the TSL2591 sensors on the edge nodes recorded extreme environmental irradiance. Initially, the Master Hub utilized this data to continuously actuate the panels into perfectly perpendicular alignment with the sun. Simultaneously, the Solar Guardian software logged the physical power generation (Voltage and Current) injected into the 12V battery bank.

When analyzing the temporal datasets over this high-intensity period, the telemetry revealed both the system's peak capabilities and the severe thermodynamic penalty of tracking desynchronization:

1. **Peak Synchronization (12:10 PM - 12:35 PM):** A direct, proportional correlation was observed between the peak irradiance detected by the digital sensors and the peak wattage ingested by the MPPT. The system demonstrated a rapid stabilization of power, climbing from 7.66 W at 12:10 PM to consistently outputting over 20 W by 12:15 PM, eventually culminating in a peak yield of 23.33 W at 12:25 PM. By maintaining a perfect perpendicular angle, the dual-axis tracker successfully mitigated cosine losses, sustaining a near-maximum theoretical power generation plateau.

2. Transient Sensor Anomaly & Cosine Loss Demonstration (12:35 PM - 12:40 PM):

At approximately 12:35 PM, the system experienced a transient sensor desynchronization event. Erroneous differential readings caused the Master Hub to drive the linear actuators off-axis, physically misaligning the solar array from the sun's actual azimuth. The Solar Guardian MPPT data immediately recorded a sharp, distinct drop in harvested energy during this 5-minute window, plummeting from a high of 21.45 W down to just 11.99 W. Rather than a system failure, this anomaly provided excellent empirical validation of the physics behind solar tracking. The moment the panel deviated from a perpendicular state, severe cosine losses were incurred, proving that continuous, high-precision actuation is critical for optimal yield.

Solar Guardian MPPT Electrical Yield Data (12:10 PM - 1:00 PM)

Time	PV Information- PV Power (W)
12:10:00	7.6648
12:11:00	7.2324
12:12:00	13.4702
12:13:00	11.5596
12:14:00	10.5506
12:15:00	20.55
12:16:00	15.2004
12:17:00	19.3094
12:18:00	20.355
12:19:00	20.2488
12:20:00	13.5369
12:21:00	21.1968
12:25:00	23.332
12:30:00	22.801
12:35:00	21.456
12:40:00	11.992
12:45:00	13.3641
12:50:00	16.2064
12:55:00	17.2312
13:00:00	3.9636

Table 9.1 details the granular electrical generation metrics logged during this period, highlighting the power dip during the anomaly.

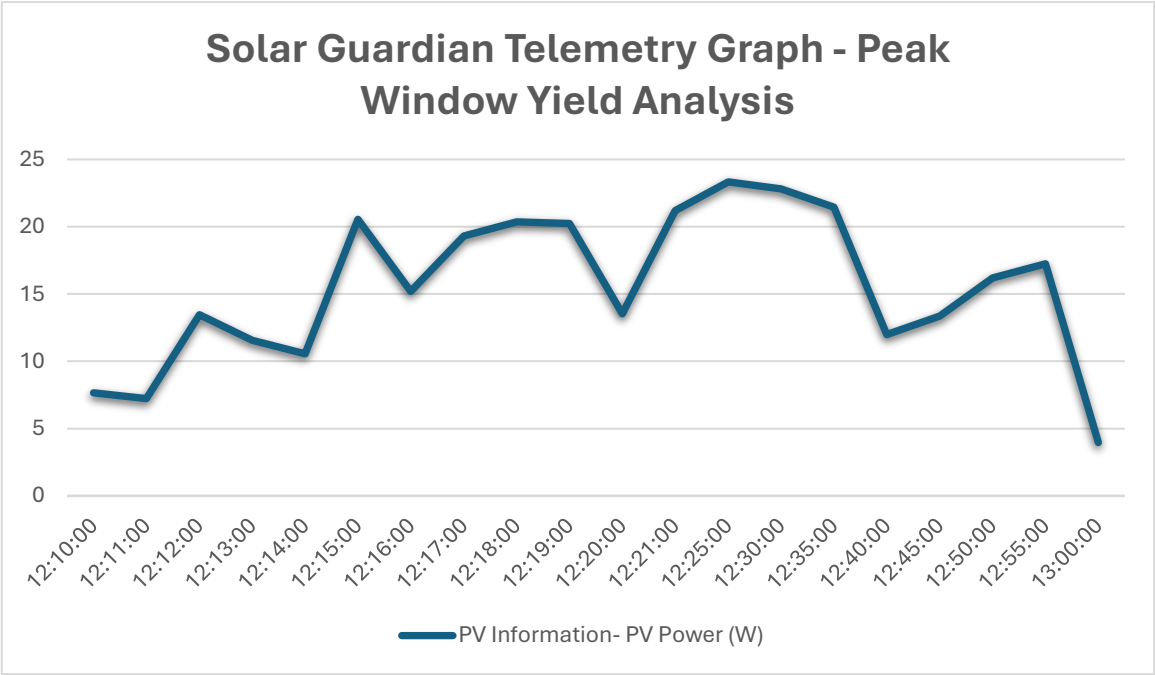


Figure 9.1 visually correlates the MPPT yield curve against the tracker's physical positioning).

9.2 WEATHER DEGRADATION AND AI STOWING IMPACT

The sudden termination of the clear-sky window at 1:00 PM provided a critical stress test for the Soltra ecosystem's Edge AI layer. As heavy, diffuse storm clouds rapidly occluded the sun, direct irradiance levels plummeted, and the MPPT generation dropped significantly, collapsing to a marginal 3.96 W.

In a traditional tracking system utilizing primitive LDR baffles, this sudden transition to diffuse, scattered light would trigger catastrophic "cloud hunting." The motor drivers would wildly oscillate the heavy panels back and forth, consuming excessive battery power in a futile attempt to locate a non-existent direct light source.

However, the telemetry confirmed the architectural superiority of Soltra AI integration. As the environmental lighting grew chaotic, the Roboflow object detection model and the Gemini Vision API fallback successfully classified the sky as heavily overcast. This triggered a deterministic "STOW" command. The Master Hub deliberately halted all pan/tilt tracking logic,

overriding the confused LDR sensors, and commanded the linear actuators to lay the panel completely flat.

By analyzing the Solar Guardian data from the moment the storm rolled in, it is mathematically evident that the AI override prevented a massive parasitic energy drain. The system successfully conserved the charge generated during the peak window, proving that intelligent, predictive stowing is just as critical to total net yield as accurate sun tracking.

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CHAPTER 11

REFERENCES & EXTERNAL LINKS.

[11] EasyEDA, "EasyEDA Online PCB Design & Circuit Simulator," EasyEDA. [Online].

Available: <https://easyeda.com/>

[12] KiCad, "KiCad EDA - Schematic Capture & PCB Design Software," KiCad. [Online].

Available: <https://www.kicad.org/>

Appendix D: Solar Guardian MPPT Telemetry & Raw Data

The raw telemetry logs captured via the Solar Guardian software suite. The interface connects directly to the system's MPPT charge controller, providing real-time, high-resolution data on PV Information, Battery Information, Load Information and Device Information. These logs empirically validate the dual-axis tracking efficiency and energy yield analysis discussed in Chapter 9.

